



National  
Qualifications  
2015

---

# **2015 Mathematics**

## **New Higher Paper 1**

### **Finalised Marking Instructions**

© Scottish Qualifications Authority 2015

The information in this publication may be reproduced to support SQA qualifications only on a non-commercial basis. If it is to be used for any other purposes written permission must be obtained from SQA's NQ Assessment team.

Where the publication includes materials from sources other than SQA (secondary copyright), this material should only be reproduced for the purposes of examination or assessment. If it needs to be reproduced for any other purpose it is the centre's responsibility to obtain the necessary copyright clearance. SQA's NQ Assessment team may be able to direct you to the secondary sources.

These Marking Instructions have been prepared by Examination Teams for use by SQA Appointed Markers when marking External Course Assessments. This publication must not be reproduced for commercial or trade purposes.



## General Comments

These marking instructions are for use with the 2015 Higher Mathematics Examination.

For each question the marking instructions are in two sections, namely **Illustrative Scheme** and **Generic Scheme**. The **Illustrative Scheme** covers methods which are commonly seen throughout the marking. The **Generic Scheme** indicates the rationale for which each mark is awarded. In general, markers should use the **Illustrative Scheme** and only use the **Generic Scheme** where a candidate has used a method not covered in the **Illustrative Scheme**.

All markers should apply the following general marking principles throughout their marking:

- 1 Marks must be assigned in accordance with these marking instructions. In principle, marks are awarded for what is correct, rather than deducted for what is wrong.
- 2 One mark is available for each •. There are **no** half marks.
- 3 Working subsequent to an error **must be followed through**, with possible full marks for the subsequent working, provided that the level of difficulty involved is approximately similar. Where, subsequent to an error, the working for a follow through mark has been eased, the follow through mark cannot be awarded.
- 4 As indicated on the front of the question paper, full credit should only be given where the solution contains appropriate working. Throughout this paper, unless specifically mentioned in the marking instructions, a correct answer with no working receives no credit.
- 5 In general, as a consequence of an error perceived to be trivial, casual or insignificant, e.g.  $6 \times 6 = 12$ , candidates lose the opportunity of gaining a mark. But note the second example in comment 7.
- 6 Where a transcription error (paper to script or within script) occurs, the candidate should be penalised, eg

The diagram illustrates three scenarios for marking a transcription error in a quadratic equation. Each scenario is shown in a box with a callout explaining the marking decision.

**Scenario 1:** The box shows the equations  $x^2 + 5x + 7 = 9x + 4$ ,  $x - 4x + 3 = 0$ , and  $x = 1$ . The first equation is marked with a red checkmark. The second equation is marked with a red X. The third equation is marked with a red checkmark and a '2' in a box. A callout box points to the second equation, stating: "This is a transcription error and so the mark is not awarded."

**Scenario 2:** The box shows the equations  $x^2 + 5x + 7 = 9x + 4$ ,  $x - 4x + 3 = 0$ , and  $x = 1$ . The first equation is marked with a red checkmark. The second equation is marked with a red X. The third equation is marked with a red checkmark and a '2' in a box. A callout box points to the second equation, stating: "Eased as no longer a solution of a quadratic equation."

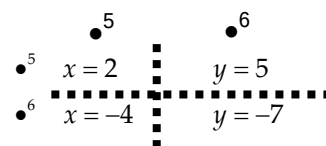
**Scenario 3:** The box shows the equations  $x^2 + 5x + 7 = 9x + 4$ ,  $x - 4x + 3 = 0$ , and  $(x - 3)(x - 1) = 0$ . The first equation is marked with a red checkmark. The second equation is marked with a red checkmark. The third equation is marked with a red checkmark. A callout box points to the second equation, stating: "Exceptionally this error is not treated as a transcription error as the candidate deals with the intended quadratic equation. The candidate has been given the benefit of the doubt."

## 7 Vertical/horizontal marking

Where a question results in two pairs of solutions, this technique should be applied, but only if indicated in the detailed marking instructions for the question.

Example: Point of intersection of line with curve

Illustrative Scheme: •<sup>5</sup>  $x = 2, x = -4$   
•<sup>6</sup>  $y = 5, y = -7$



Markers should choose whichever method benefits the candidate, but **not** a combination of both.

- 8 In final answers, numerical values should be simplified as far as possible, unless specifically mentioned in the detailed marking instructions.

Examples:  $\frac{15}{12}$  should be simplified to  $\frac{5}{4}$  or  $1\frac{1}{4}$        $\frac{43}{1}$  should be simplified to 43

$\frac{15}{0.3}$  should be simplified to 50

$\frac{4/5}{3}$  should be simplified to  $\frac{4}{15}$

$\sqrt{64}$  must be simplified to 8

The square root of perfect squares up to and including 100 must be known.

- 9 Commonly Observed Responses (COR) are shown in the marking instructions to help mark common and/or non-routine solutions. CORs may also be used as a guide when marking similar non-routine candidate responses.
- 10 Unless specifically mentioned in the marking instructions, the following should not be penalised:
- Working subsequent to a **correct** answer;
  - Correct working in the wrong part of a question;
  - Legitimate variations in numerical answers, eg angles in degrees rounded to nearest degree;
  - Omission of units;
  - Bad form (bad form only becomes bad form if subsequent working is correct), e.g.  
 $(x^3 + 2x^2 + 3x + 2)(2x + 1)$   
written as  
 $(x^3 + 2x^2 + 3x + 2) \times 2x + 1$   
 $2x^4 + 4x^3 + 6x^2 + 4x + x^3 + 2x^2 + 3x + 2$   
 $2x^4 + 5x^3 + 8x^2 + 7x + 2$  gains full credit;
  - Repeated error within a question, but not between questions.
- 11 In any 'Show that . . .' question, where the candidate has to arrive at a required result, the last mark of that part is not available as a follow through from a previous error unless specifically stated otherwise in the detailed marking instructions.

- 12** All working should be carefully checked, even where a fundamental misunderstanding is apparent early in the candidate's response. Marks may still be available later in the question so reference must be made continually to the marking instructions.  
All working must be checked: the appearance of the correct answer does not necessarily indicate that the candidate has gained all the available marks.
- 13** If you are in serious doubt whether a mark should or should not be awarded, consult your Team Leader (TL).
- 14** Scored out working which **has not been replaced** should be marked where still legible. However, if the scored out working **has been replaced**, only the work which has not been scored out should be marked.
- 15** Where a candidate has made multiple attempts using the same strategy, mark all attempts and award the lowest mark.  
Where a candidate has tried different strategies, apply the above ruling to attempts within each strategy and then award the highest resultant mark. For example:

|                                                                    |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| Strategy 1 attempt 1 is worth 3 marks                              | Strategy 2 attempt 1 is worth 1 mark                               |
| Strategy 1 attempt 2 is worth 4 marks                              | Strategy 2 attempt 2 is worth 5 marks                              |
| From the attempts using strategy 1, the resultant mark would be 3. | From the attempts using strategy 2, the resultant mark would be 1. |

In this case, award 3 marks.

- 16** In cases of difficulty, covered neither in detail nor in principle in these instructions, markers should contact their TL in the first instance.

## Detailed Marking Instructions for each question

| Question                                                                                                                                                                                                                         |  | Generic Scheme                                                                                                                                                                                                                                     | Illustrative Scheme                                                                                                                                                                     | Max Mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>1.</b>                                                                                                                                                                                                                        |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |          |
|                                                                                                                                                                                                                                  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> equate scalar product to zero</li> <li>•<sup>2</sup> state value of <math>t</math></li> </ul>                                                                                                 | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>-24 + 2t + 6 = 0</math></li> <li>•<sup>2</sup> <math>t = 9</math></li> </ul>                                                 | <b>2</b> |
| <b>Notes:</b>                                                                                                                                                                                                                    |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                              |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |          |
| <b>Candidate A</b><br>$-24 + 2t + 6 = -1$ • <sup>1</sup> <span style="color: red;">×</span><br>$t = \frac{17}{2}$ or $8\frac{1}{2}$ • <sup>2</sup> <span style="border: 1px solid red; padding: 0 2px;">✓1</span>                |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |          |
| <b>2.</b>                                                                                                                                                                                                                        |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |          |
|                                                                                                                                                                                                                                  |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> know to and differentiate</li> <li>•<sup>2</sup> evaluate <math>\frac{dy}{dx}</math></li> <li>•<sup>3</sup> evaluate y-coordinate</li> <li>•<sup>4</sup> state equation of tangent</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>6x^2</math></li> <li>•<sup>2</sup> 24</li> <li>•<sup>3</sup> -13</li> <li>•<sup>4</sup> <math>y = 24x + 35</math></li> </ul> | <b>4</b> |
| <b>Notes:</b>                                                                                                                                                                                                                    |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |          |
| 1. • <sup>4</sup> is only available if an attempt has been made to find the gradient from differentiation.<br>2. At mark • <sup>4</sup> accept $y + 13 = 24(x + 2)$ , $y - 24x = 35$ or any other rearrangement of the equation. |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                              |  |                                                                                                                                                                                                                                                    |                                                                                                                                                                                         |          |

| Question | Generic Scheme                                                                                                                                                                                                                                         | Illustrative Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Max Mark |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3.       |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|          | <ul style="list-style-type: none"> <li>•<sup>1</sup> know to use <math>x = -3</math></li> <li>•<sup>2</sup> interpret result and state conclusion</li> <li>•<sup>3</sup> state quadratic factor</li> <li>•<sup>4</sup> factorise completely</li> </ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(-3)^3 - 3(-3)^2 - 10(-3) + 24</math></li> <li>•<sup>2</sup> <math>= 0 \therefore (x + 3)</math> is a factor.</li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math display="block">\begin{array}{r rrrr} -3 &amp; 1 &amp; -3 &amp; -10 &amp; 24 \\ &amp; &amp; -3 &amp; &amp; \\ \hline &amp; 1 &amp; &amp; &amp; \end{array}</math> </li> <li>•<sup>2</sup> <math display="block">\begin{array}{r rrrr} -3 &amp; 1 &amp; -3 &amp; -10 &amp; 24 \\ &amp; &amp; -3 &amp; 18 &amp; -24 \\ \hline &amp; 1 &amp; -6 &amp; 8 &amp; 0 \end{array}</math> <p>remainder = 0 <math>\therefore (x + 3)</math> is a factor.</p> </li> </ul> <p><b>Method 3</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math display="block">\begin{array}{r} x^2 \\ x+3 \overline{) x^3 - 3x^2 - 10x + 24} \\ \underline{x^3 + 3x^2} \phantom{- 10x + 24} \end{array}</math> </li> <li>•<sup>2</sup> <math>= 0 \therefore (x + 3)</math> is a factor.</li> <li>•<sup>3</sup> <math>x^2 - 6x + 8</math> stated or implied by •<sup>4</sup></li> <li>•<sup>4</sup> <math>(x + 3)(x - 4)(x - 2)</math></li> </ul> | 4        |

#### Notes:

- Communication at •<sup>2</sup> must be consistent with working at that stage ie a candidate's working must arrive legitimately at 0 before •<sup>2</sup> is awarded.
- Accept any of the following for •<sup>2</sup>:  
' $f(-3) = 0$  so  $(x + 3)$  is a factor'  
'since remainder is 0, it is a factor'  
the 0 from the table linked to the word 'factor' by eg 'so', 'hence', ' $\therefore$ ', ' $\rightarrow$ ', ' $\Rightarrow$ '
- Do not accept any of the following for •<sup>2</sup>:  
double underlining the zero or boxing the zero without comment  
' $x = 3$  is a factor', ' $(x - 3)$  is a factor', ' $x = -3$  is a root', ' $(x - 3)$  is a root', " $(x + 3)$  is a root"  
the word 'factor' **only**, with no link
- At •<sup>4</sup> the expression may be written in any order.
- An incorrect quadratic correctly factorised may gain •<sup>4</sup>
- Where the quadratic factor obtained is irreducible, candidates must clearly demonstrate that  $b^2 - 4ac < 0$  to gain •<sup>4</sup>
- $= 0$  must appear at •<sup>1</sup> or •<sup>2</sup> for •<sup>2</sup> to be awarded.
- For candidates who do not arrive at 0 at the •<sup>2</sup> stage •<sup>2</sup>•<sup>3</sup>•<sup>4</sup> not available.
- Do not penalise candidates who attempt to solve a cubic equation. However, within this working there may be evidence of the correct factorisation of the cubic.

| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Candidate A                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                  | Candidate B                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| $\begin{array}{r} 2 \quad 1 \quad -3 \quad -10 \quad 24 \\ \quad \quad 2 \quad -2 \quad -24 \\ \hline 1 \quad -1 \quad -12 \quad 0 \end{array} \Rightarrow x-2 \text{ is a factor}$ $(x-2)(x^2-x-12)$ $(x-2)(x-4)(x+3) \Rightarrow x+3 \text{ is a factor}$ <div><div>•<sup>2</sup> ✓1</div><div>•<sup>3</sup> ✓</div><div>•<sup>4</sup> ✓</div><div>•<sup>1</sup> ✓</div></div> |                                                                                                                                                                                                                                                                                                  | $\begin{array}{r} 2 \quad 1 \quad -3 \quad -10 \quad 24 \\ \quad \quad 2 \quad -2 \quad -24 \\ \hline 1 \quad -1 \quad -12 \quad 0 \end{array} \Rightarrow x-2 \text{ is a factor}$ <div><div>•<sup>1</sup> ✓2</div><div>•<sup>2</sup> ✓1</div></div>                                                                                                                                                                                                                      |   |
| 4.                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
|                                                                                                                                                                                                                                                                                                                                                                                  | <div>•<sup>1</sup> state the value of <math>p</math></div> <div>•<sup>2</sup> state the value of <math>q</math></div> <div>•<sup>3</sup> state the value of <math>r</math></div>                                                                                                                 | <div>•<sup>1</sup> <math>p = 3</math></div> <div>•<sup>2</sup> <math>q = 4</math></div> <div>•<sup>3</sup> <math>r = 1</math></div>                                                                                                                                                                                                                                                                                                                                        | 3 |
| Notes:                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| 1. These are the only acceptable responses for $p$ , $q$ and $r$ .                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| 5(a).                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
|                                                                                                                                                                                                                                                                                                                                                                                  | <div>•<sup>1</sup> let <math>y = 6 - 2x</math> and rearrange.</div> <div>•<sup>2</sup> state expression.</div> <div>Method 2</div> <div>•<sup>3</sup> equates composite function to <math>x</math></div> <div>•<sup>1</sup> start to rearrange.</div> <div>•<sup>2</sup> state expression.</div> | <div>•<sup>1</sup> <math>x = \frac{6-y}{2}</math> or <math>y = \frac{6-x}{2}</math></div> <div>•<sup>2</sup> <math>g^{-1}(x) = \frac{6-x}{2}</math> or <math>3 - \frac{x}{2}</math> or <math>\frac{x-6}{-2}</math></div> <div>Method 2</div> <div><math>g(g^{-1}(x)) = x</math> this gains •<sup>3</sup></div> <div><math>6 - 2g^{-1}(x) = x</math></div> <div><math>g^{-1}(x) = \frac{6-x}{2}</math> or <math>3 - \frac{x}{2}</math> or <math>\frac{x-6}{-2}</math></div> | 2 |
| Notes:                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| 1. At • <sup>1</sup> accept any equivalent expression with any 2 distinct variables.                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| 5(b).                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
|                                                                                                                                                                                                                                                                                                                                                                                  | • <sup>3</sup> state expression                                                                                                                                                                                                                                                                  | • <sup>3</sup> $x$                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 |
| Notes:                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| 2. Candidates using method 2 may be awarded • <sup>3</sup> at line one.                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| 3. For candidates who attempt to find the composite function $g(g^{-1}(x))$ , accept $6 - 2\left(\frac{6-x}{2}\right)$ for • <sup>3</sup> .                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| 4. In this case • <sup>3</sup> may be awarded as follow through where an incorrect $g^{-1}(x)$ is found at • <sup>2</sup> , provided it includes the variable $x$ .                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                |  | Generic Scheme                                                                                                                                                                                                                                                 | Illustrative Scheme                                                                                                                                                                                                                                                                 | Max Mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 6.                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> use laws of logs</li> <li>•<sup>2</sup> use laws of logs</li> <li>•<sup>3</sup> evaluate log</li> </ul>                                                                                                   | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\log_6 27^{\frac{1}{3}}</math></li> <li>•<sup>2</sup> <math>\log_6 \left( 12 \times 27^{\frac{1}{3}} \right)</math></li> <li>•<sup>3</sup> 2</li> </ul>                                                                  | 3        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| Candidate A                                                                                                                                                                                                                                                                                                                                                                                                                             |  | Candidate B                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                     |          |
| $\log_6 12 + \log_6 9$                                                                                                                                                                                                                                                                                                                                                                                                                  |  | $\frac{1}{3} \log_6 (12 \times 27)$                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                     |          |
| $\log_6 (12 \times 9)$                                                                                                                                                                                                                                                                                                                                                                                                                  |  | $\frac{1}{3} \log_6 324$                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                     |          |
| $\log_6 108$                                                                                                                                                                                                                                                                                                                                                                                                                            |  | $\log_6 324^{\frac{1}{3}}$                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                     |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | Award 1 out of 3 ^,^ <input checked="" type="checkbox"/> 1                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                     |          |
| 7.                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> write in differentiable form</li> <li>•<sup>2</sup> differentiate first term</li> <li>•<sup>3</sup> differentiate second term</li> <li>•<sup>4</sup> evaluate derivative at <math>x = 4</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>3x^{\frac{3}{2}} - 2x^{-1}</math></li> <li>•<sup>2</sup> <math>\frac{9}{2}x^{\frac{1}{2}} + \dots</math></li> <li>•<sup>3</sup> <math>\dots + 2x^{-2}</math></li> <li>•<sup>4</sup> <math>9\frac{1}{8}</math></li> </ul> | 4        |
| Notes:                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| 1. • <sup>2</sup> must involve a fractional index.<br>2. • <sup>3</sup> must involve a negative index.<br>3. • <sup>4</sup> is only available as a consequence of substituting into a 'derivative' containing a fractional or negative index.<br>4. If no attempt has been made to expand the bracket at • <sup>1</sup> then • <sup>2</sup> & • <sup>3</sup> are not available.<br>• <sup>4</sup> is still available as follow through. |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| Commonly Observed Responses:                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| Candidate A                                                                                                                                                                                                                                                                                                                                                                                                                             |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| $f(x) = 3x^{\frac{1}{2}} - 2x^{-\frac{1}{4}}$                                                                                                                                                                                                                                                                                                                                                                                           |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| $f'(x) = \frac{3}{2}x^{-\frac{1}{2}} + \frac{1}{2}x^{-\frac{5}{4}}$                                                                                                                                                                                                                                                                                                                                                                     |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| $= \frac{3}{2\sqrt{x}} + \frac{1}{2\sqrt[4]{x^5}}$                                                                                                                                                                                                                                                                                                                                                                                      |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| $f'(4) = \frac{3}{2\sqrt{4}} + \frac{1}{2\sqrt[4]{4^5}}$                                                                                                                                                                                                                                                                                                                                                                                |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |
| $= \frac{3}{4} + \frac{1}{8\sqrt{2}}$                                                                                                                                                                                                                                                                                                                                                                                                   |  |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                     |          |



| Question                                                                                      |  | Generic Scheme                                                                                                                                                                                                  | Illustrative Scheme                                                                                                                                                                                                                                        | Max Mark                                |
|-----------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 8.                                                                                            |  |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                            |                                         |
|                                                                                               |  | <ul style="list-style-type: none"> <li><sup>1</sup> interpret information</li> <li><sup>2</sup> express in standard quadratic form</li> <li><sup>3</sup> factorise</li> <li><sup>4</sup> state range</li> </ul> | <ul style="list-style-type: none"> <li><sup>1</sup> <math>x(x-2) &lt; 15</math></li> <li><sup>2</sup> <math>x^2 - 2x - 15 &lt; 0</math></li> <li><sup>3</sup> <math>(x-5)(x+3) &lt; 0</math></li> <li><sup>4</sup> <math>2 &lt; x &lt; 5</math></li> </ul> | 4                                       |
| Notes:                                                                                        |  |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                            |                                         |
| Commonly Observed Responses:                                                                  |  |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                            |                                         |
| <b>Candidate A</b><br>$x(x-2) = 15$<br>$x^2 - 2x - 15 = 0$<br>$x = -3, 5$                     |  | <ul style="list-style-type: none"> <li><sup>1</sup> ✗</li> <li><sup>2</sup> ✓ 2</li> <li><sup>3</sup> ✓ 1</li> <li><sup>4</sup> ^</li> </ul>                                                                    | <b>Candidate B - Mistaking perimeter for area</b><br>$4x - 4 < 15$<br>$x < \frac{19}{4}$<br><b>Award 1/4</b>                                                                                                                                               |                                         |
| <b>Candidate C</b><br>$x^2 - 2x < 15$<br>$x > 2$<br><b>Award 1/4</b>                          |  |                                                                                                                                                                                                                 | <b>Candidate D</b><br>$x^2 - 2x < 15$<br>$x > 2$<br>$x < 5$<br><b>Award 2/4</b>                                                                                                                                                                            | <b>Inequalities not linked by 'and'</b> |
| <b>Candidate E</b><br>$x^2 - 2x < 15$<br>$x > 2$<br><b>and</b><br>$x < 5$<br><b>Award 4/4</b> |  | <b>Inequalities linked by 'and'</b>                                                                                                                                                                             |                                                                                                                                                                                                                                                            |                                         |

| Question                                                                                                                                                                                                                                                                                |  | Generic Scheme                                                                                                                                                                                                                                                                                                                             | Illustrative Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Max Mark |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 9.                                                                                                                                                                                                                                                                                      |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                                                                                                                                                                                                                                                                                         |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> find gradient of AB</li> <li>•<sup>2</sup> calculate gradient of BC</li> <li>•<sup>3</sup> interpret results and state conclusion</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>m_{AB} = -\sqrt{3}</math></li> <li>•<sup>2</sup> <math>m_{BC} = -\frac{1}{\sqrt{3}}</math></li> <li>•<sup>3</sup> <math>m_{AB} \neq m_{BC} \Rightarrow</math> points are not collinear.</li> </ul> <p style="text-align: center;"><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>m_{AB} = -\sqrt{3}</math></li> <li>•<sup>2</sup> AB makes <math>120^\circ</math> with positive direction of the <math>x</math>-axis.</li> <li>•<sup>3</sup> <math>120 \neq 150</math> so points are not collinear.</li> </ul> | <b>3</b> |
| <b>Notes:</b>                                                                                                                                                                                                                                                                           |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| 1. The statement made at • <sup>3</sup> must be consistent with the gradients or angles found for • <sup>1</sup> and • <sup>2</sup> .                                                                                                                                                   |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                     |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                                                                                                                                                                                                                                                                                         |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| 10(a).                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                                                                                                                                                                                                                                                                                         |  | • <sup>1</sup> state value of $\cos 2x$                                                                                                                                                                                                                                                                                                    | • <sup>1</sup> $\frac{4}{5}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>1</b> |
| <b>Notes:</b>                                                                                                                                                                                                                                                                           |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                                                                                                                                                                                                                                                                                         |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                     |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Candidate A</b>                                                                                                                                                                                                                                                                      |  | <b>Candidate B</b>                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| $\cos 2x = \frac{3}{5}$ <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\times</math></li> <li>•<sup>2</sup> <input checked="" type="checkbox"/></li> <li>•<sup>3</sup> <input checked="" type="checkbox"/></li> </ul> $2\cos^2 x - 1 = \dots$ $\cos x = \frac{2}{\sqrt{5}}$ |  | $\cos 2x = 4$ <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\times</math></li> <li>•<sup>2</sup> <input checked="" type="checkbox"/></li> </ul> $2\cos^2 x - 1 = 4$ $\cos^2 x = \frac{5}{2}$ $\cos x = \sqrt{\frac{5}{2}}$ <ul style="list-style-type: none"> <li>•<sup>3</sup> <math>\times</math> invalid answer</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| 10(b).                                                                                                                                                                                                                                                                                  |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                                                                                                                                                                                                                                                                                         |  | <ul style="list-style-type: none"> <li>•<sup>2</sup> use double angle formula</li> <li>•<sup>3</sup> evaluate <math>\cos x</math></li> </ul>                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>•<sup>2</sup> <math>2\cos^2 x - 1 = \dots</math></li> <li>•<sup>3</sup> <math>\frac{3}{\sqrt{10}}</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>2</b> |
| <b>Notes:</b>                                                                                                                                                                                                                                                                           |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| 1. Ignore the inclusion of $-\frac{3}{\sqrt{10}}$ .<br>2. At • <sup>2</sup> the double angle formula must be equated to the candidates answer to part (a).                                                                                                                              |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                     |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                                                                                                                                                                                                                                                                                         |  |                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |

| Question                                                                                                                                                                                                                                                                               |  | Generic Scheme                                                                                                                                                                                                                                      | Illustrative Scheme                                                                                                                                                                                            | Max Mark |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11(a).                                                                                                                                                                                                                                                                                 |  |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                |          |
|                                                                                                                                                                                                                                                                                        |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> state coordinates of centre</li> <li>•<sup>2</sup> find gradient of radius</li> <li>•<sup>3</sup> state perpendicular gradient</li> <li>•<sup>4</sup> determine equation of tangent</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>(-8, -2)</math></li> <li>•<sup>2</sup> <math>-\frac{1}{2}</math></li> <li>•<sup>3</sup> 2</li> <li>•<sup>4</sup> <math>y = 2x - 1</math></li> </ul> | 4        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                          |  |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                |          |
| <p>1. •<sup>4</sup> is only available as a consequence of trying to find and use a perpendicular gradient.</p> <p>2. At mark •<sup>4</sup> accept <math>y + 5 = 2(x + 2)</math>, <math>y - 2x = -1</math>, <math>y - 2x + 1 = 0</math> or any other rearrangement of the equation.</p> |  |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                    |  |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                |          |
|                                                                                                                                                                                                                                                                                        |  |                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                |          |

| Question | Generic Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Illustrative Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Max Mark |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 11(b).   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
|          | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>5</sup> arrange equation of tangent in appropriate form and equate <math>y_{\text{tangent}}</math> to <math>y_{\text{parabola}}</math></li> <li>•<sup>6</sup> rearrange and equate to 0</li> <li>•<sup>7</sup> know to use discriminant and identify <math>a</math>, <math>b</math>, and <math>c</math></li> <li>•<sup>8</sup> simplify and equate to 0</li> <li>•<sup>9</sup> start to solve</li> <li>•<sup>10</sup> state value of <math>p</math></li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>5</sup> arrange equation of tangent in appropriate form and equate <math>y_{\text{tangent}}</math> to <math>y_{\text{parabola}}</math></li> <li>•<sup>6</sup> find <math>\frac{dy}{dx}</math> for parabola</li> <li>•<sup>7</sup> equate to gradient of line and rearrange for <math>p</math></li> <li>•<sup>8</sup> substitute and arrange in standard form</li> <li>•<sup>9</sup> factorise and solve for <math>x</math></li> <li>•<sup>10</sup> state value of <math>p</math></li> </ul> | <p><b>Method 1</b></p> <ul style="list-style-type: none"> <li>•<sup>5</sup> <math>2x - 1 = -2x^2 + px + 1 - p</math></li> <li>•<sup>6</sup> <math>2x^2 + (2 - p)x + p - 2 = 0</math></li> <li>•<sup>7</sup> <math>(2 - p)^2 - 4 \times 2 \times (p - 2)</math></li> <li>•<sup>8</sup> <math>p^2 - 12p + 20 = 0</math></li> <li>•<sup>9</sup> <math>(p - 10)(p - 2) = 0</math></li> <li>•<sup>10</sup> <math>p = 10</math></li> </ul> <p><b>Method 2</b></p> <ul style="list-style-type: none"> <li>•<sup>5</sup> <math>2x - 1 = -2x^2 + px + 1 - p</math></li> <li>•<sup>6</sup> <math>\frac{dy}{dx} = -4x + p</math></li> <li>•<sup>7</sup> <math>2 = -4x + p</math><br/><math>p = 2 + 4x</math></li> <li>•<sup>8</sup> <math>0 = 2x^2 - 4x</math></li> <li>•<sup>9</sup> <math>0 = 2x(x - 2)</math><br/><math>x = 0, x = 4</math></li> <li>•<sup>10</sup> <math>p = 10</math></li> </ul> | 6        |

#### Notes:

1. At •<sup>6</sup> accept  $2x^2 + 2x - px + p - 2 = 0$ .
2. At •<sup>7</sup> accept  $a = 2$ ,  $b = (2 - p)$ , and  $c = (p - 2)$ .

#### Commonly Observed Responses:

##### Just using the parabola

$$a = -2 \quad b = p \quad c = 1 - p$$

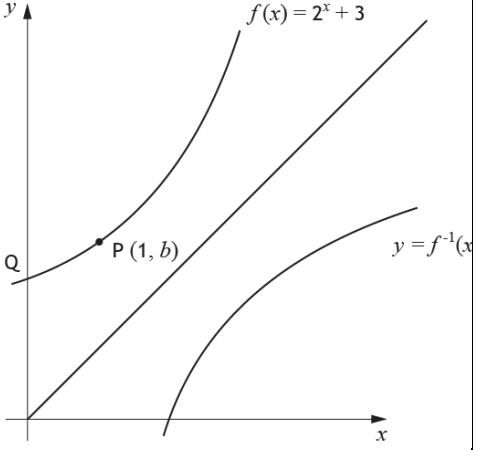
$$b^2 - 4ac = p^2 - 4 \times (-2)(1 - p)$$

$$= p^2 - 8p + 8 = 0$$

$$p = 4 \pm \sqrt{8}$$

$$p = 4 + \sqrt{8} \quad \text{as } p > 3$$

- <sup>5</sup>  $\wedge$
- <sup>6</sup>  $\wedge$
- <sup>7</sup> ☒ 1
- <sup>8</sup> ☒ 2
- <sup>9</sup> ☒ 1
- <sup>10</sup> ☒ 1

| Question                                                                    |  | Generic Scheme                                                                                                                             | Illustrative Scheme                                                                                                                                                        | Max Mark |
|-----------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>12.</b>                                                                  |  |                                                                                                                                            |                                                                                                                                                                            |          |
|                                                                             |  | <ul style="list-style-type: none"> <li><sup>1</sup> interpret integral below <math>x</math>-axis</li> <li><sup>2</sup> evaluate</li> </ul> | <ul style="list-style-type: none"> <li><sup>1</sup> <math>-1</math> (accept area below <math>x</math>-axis = 1)</li> <li><sup>2</sup> <math>-\frac{1}{2}</math></li> </ul> | <b>2</b> |
| <b>Notes:</b>                                                               |  |                                                                                                                                            |                                                                                                                                                                            |          |
| 1. For candidates who calculate the area as $\frac{3}{2}$ award 1 out of 2. |  |                                                                                                                                            |                                                                                                                                                                            |          |
| <b>Commonly Observed Responses:</b>                                         |  |                                                                                                                                            |                                                                                                                                                                            |          |
|                                                                             |  |                                                                                                                                            |                                                                                                                                                                            |          |
| <b>13(a)</b>                                                                |  |                                                                                                                                            |                                                                                                                                                                            |          |
|                                                                             |  | <ul style="list-style-type: none"> <li><sup>1</sup> calculate <math>b</math></li> </ul>                                                    | <ul style="list-style-type: none"> <li><sup>1</sup> 5</li> </ul>                                                                                                           | <b>1</b> |
| <b>Notes:</b>                                                               |  |                                                                                                                                            |                                                                                                                                                                            |          |
|                                                                             |  |                                                                                                                                            |                                                                                                                                                                            |          |
| <b>Commonly Observed Responses:</b>                                         |  |                                                                                                                                            |                                                                                                                                                                            |          |
|                                                                             |  |                                                                                                                                            |                                                                                                                                                                            |          |
| <b>13 (b)(i)</b>                                                            |  |                                                                                                                                            |                                                                                                                                                                            |          |
|                                                                             |  | <ul style="list-style-type: none"> <li><sup>2</sup> reflecting in the line <math>y = x</math></li> </ul>                                   | <ul style="list-style-type: none"> <li><sup>2</sup> </li> </ul>                         | <b>1</b> |
| <b>Notes:</b>                                                               |  |                                                                                                                                            |                                                                                                                                                                            |          |
| 1. If the reflected graph cuts the $y$ -axis, <sup>2</sup> is not awarded.  |  |                                                                                                                                            |                                                                                                                                                                            |          |
| <b>Commonly Observed Responses:</b>                                         |  |                                                                                                                                            |                                                                                                                                                                            |          |
|                                                                             |  |                                                                                                                                            |                                                                                                                                                                            |          |

| Question                                                                                                                                                                                                                                                                                                    |                    | Generic Scheme                                                                                                                                                                                                   | Illustrative Scheme                                                                                                                                                                                                                                                                                                                                   | Max Mark |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>13(b)(ii)</b>                                                                                                                                                                                                                                                                                            |                    |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                       |          |
|                                                                                                                                                                                                                                                                                                             |                    | <ul style="list-style-type: none"> <li>•<sup>3</sup> calculate <math>y</math> intercept</li> <li>•<sup>4</sup> state coordinates of image of Q</li> <li>•<sup>5</sup> state coordinates of image of P</li> </ul> | <ul style="list-style-type: none"> <li>•<sup>3</sup> 4</li> <li>•<sup>4</sup> (4, 0) see note 2</li> <li>•<sup>5</sup> (5, 1)</li> </ul>                                                                                                                                                                                                              | <b>3</b> |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                               |                    |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                       |          |
| 2. • <sup>4</sup> can only be awarded if (4,0) is clearly identified either by their labelling or by their diagram.<br>3. • <sup>3</sup> is awarded for the appearance of 4, or (4,0) or (0,4) .<br>4. • <sup>5</sup> is awarded for the appearance of (5,1) . Ignore any labelling attached to this point. |                    |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                         |                    |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Candidate A</b>                                                                                                                                                                                                                                                                                          |                    | <b>Candidate B</b>                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                       |          |
| $y = f(x)$ reflected in $x$ – axis                                                                                                                                                                                                                                                                          |                    | $y = f(x)$ reflected in $y$ – axis                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                       |          |
| 4                                                                                                                                                                                                                                                                                                           | • <sup>3</sup> ✓   | 4                                                                                                                                                                                                                | • <sup>3</sup> ✓                                                                                                                                                                                                                                                                                                                                      |          |
| (0,-4)                                                                                                                                                                                                                                                                                                      | • <sup>4</sup> ✓ 2 | (0,4)                                                                                                                                                                                                            | • <sup>4</sup> ✓ 2                                                                                                                                                                                                                                                                                                                                    |          |
| (1,-5)                                                                                                                                                                                                                                                                                                      | • <sup>5</sup> ✓ 1 | (-1,5)                                                                                                                                                                                                           | • <sup>5</sup> ✓ 2                                                                                                                                                                                                                                                                                                                                    |          |
| <b>13(c)</b>                                                                                                                                                                                                                                                                                                |                    |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                       |          |
|                                                                                                                                                                                                                                                                                                             |                    | <ul style="list-style-type: none"> <li>•<sup>6</sup> state <math>x</math> coordinate of R</li> <li>•<sup>7</sup> state <math>y</math> coordinate of R</li> </ul>                                                 | <ul style="list-style-type: none"> <li>•<sup>6</sup> <math>x = 2</math></li> <li>•<sup>7</sup> <math>y = -7</math></li> </ul>                                                                                                                                                                                                                         | <b>2</b> |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                               |                    |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                                         |                    |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>14.</b>                                                                                                                                                                                                                                                                                                  |                    |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                       |          |
|                                                                                                                                                                                                                                                                                                             |                    | <ul style="list-style-type: none"> <li>•<sup>1</sup> identify length of radius</li> <li>•<sup>2</sup> determine value of <math>k</math></li> </ul>                                                               | <div> <div> <math>y</math> – axis<br/>           tangent to circle         </div> <div>           Circle passes through origin         </div> </div><br><ul style="list-style-type: none"> <li>•<sup>1</sup> <math>r = 6</math></li> <li>•<sup>2</sup> <math>k = 25</math></li> </ul> <div> <math>r = \sqrt{61}</math><br/> <math>k = 0</math> </div> | <b>2</b> |

| Question                                                                                                                                                                                                                                                                              |  | Generic Scheme                                                                                                                                                                                                                                                                                                                           | Illustrative Scheme                                                                                                                                                                                                                                                                                                                                                                                                  | Max Mark |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 15.                                                                                                                                                                                                                                                                                   |  |                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|                                                                                                                                                                                                                                                                                       |  | <ul style="list-style-type: none"> <li>•<sup>1</sup> know to integrate</li> <li>•<sup>2</sup> integrate a term</li> <li>•<sup>3</sup> complete integration</li> <li>•<sup>4</sup> find constant of integration</li> <li>•<sup>5</sup> find value of <math>k</math></li> <li>•<sup>6</sup> state expression for <math>T</math></li> </ul> | <ul style="list-style-type: none"> <li>•<sup>1</sup> <math>\int</math></li> <li>•<sup>2</sup> <math>\frac{1}{50}t^2 \dots</math> or <math>\dots - kt</math></li> <li>•<sup>3</sup> <math>\dots - kt</math> or <math>\frac{1}{50}t^2 \dots</math></li> <li>•<sup>4</sup> <math>c = 100</math></li> <li>•<sup>5</sup> <math>k = 2</math></li> <li>•<sup>6</sup> <math>T = \frac{1}{50}t^2 - 2t + 100</math></li> </ul> | 6        |
| <b>Notes:</b>                                                                                                                                                                                                                                                                         |  |                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| 1. Accept unsimplified expressions at • <sup>2</sup> and • <sup>3</sup> stage.<br>2. • <sup>4</sup> , • <sup>5</sup> and • <sup>6</sup> are not available for candidates who have not considered the constant of integration.<br>3. • <sup>1</sup> may be implied by • <sup>2</sup> . |  |                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>Commonly Observed Responses:</b>                                                                                                                                                                                                                                                   |  |                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|                                                                                                                                                                                                                                                                                       |  |                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                      |          |

[END OF MARKING INSTRUCTIONS]